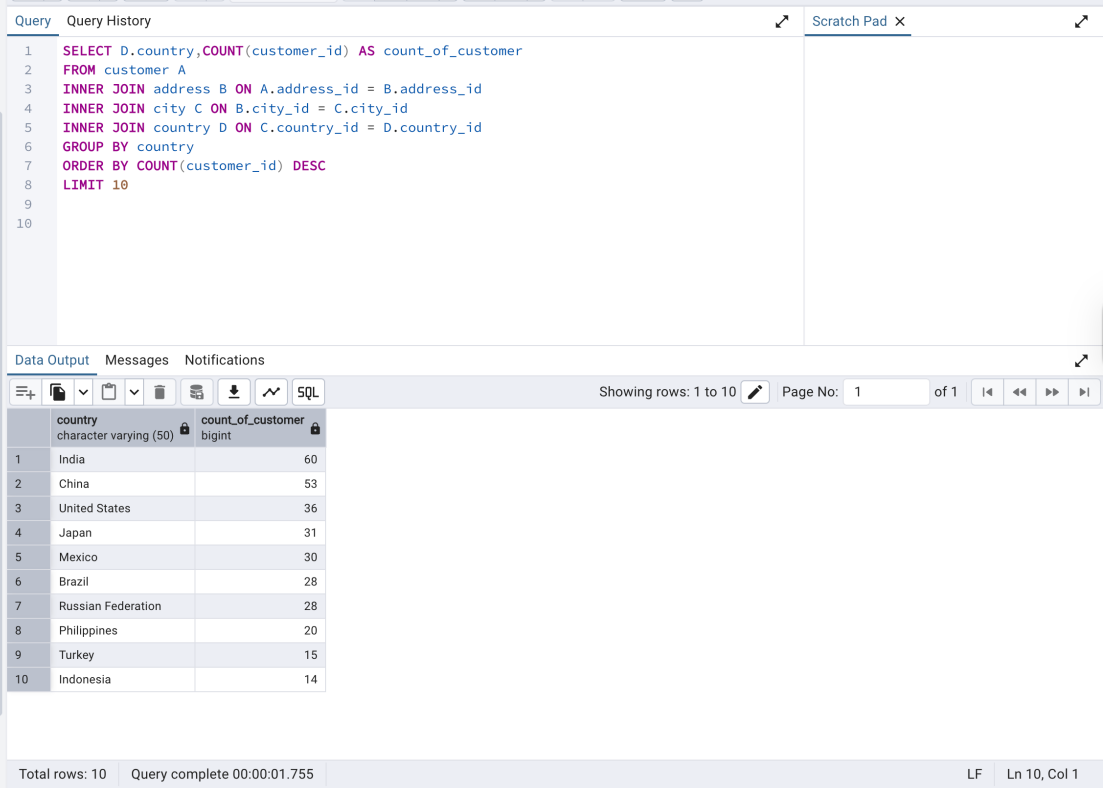


Exploratory Data Analysis

Finding the top 10 countries for Rockbuster in terms of customer numbers.

```
SELECT D.country,COUNT(customer_id) AS count_of_customer
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
GROUP BY country
ORDER BY COUNT(customer_id) DESC
LIMIT 10
```



The screenshot displays a SQL query editor with the following code:

```
1 SELECT D.country,COUNT(customer_id) AS count_of_customer
2 FROM customer A
3 INNER JOIN address B ON A.address_id = B.address_id
4 INNER JOIN city C ON B.city_id = C.city_id
5 INNER JOIN country D ON C.country_id = D.country_id
6 GROUP BY country
7 ORDER BY COUNT(customer_id) DESC
8 LIMIT 10
9
10
```

Below the query editor, the results are displayed in a table with the following columns: country (character varying (50)) and count_of_customer (bigint). The table shows the top 10 countries by customer count.

country	count_of_customer
India	60
China	53
United States	36
Japan	31
Mexico	30
Brazil	28
Russian Federation	28
Philippines	20
Turkey	15
Indonesia	14

The interface also shows a status bar at the bottom indicating "Total rows: 10" and "Query complete 00:00:01.755".

I first decided to find the concerned tables for this query and noted down their relationship with each other. The concerned tables are customer, address, city and country. Although I want the information from only two tables (customer and country), I cannot because they are not directly linked with a key. Hence, I used an inner join to establish a relationship between the tables via primary/foreign keys. The tables are connected as follows:

Customer (address_id) → Address (address_id)

Address (city_id) → City (city_id)

City (country_id) → Country (country_id)

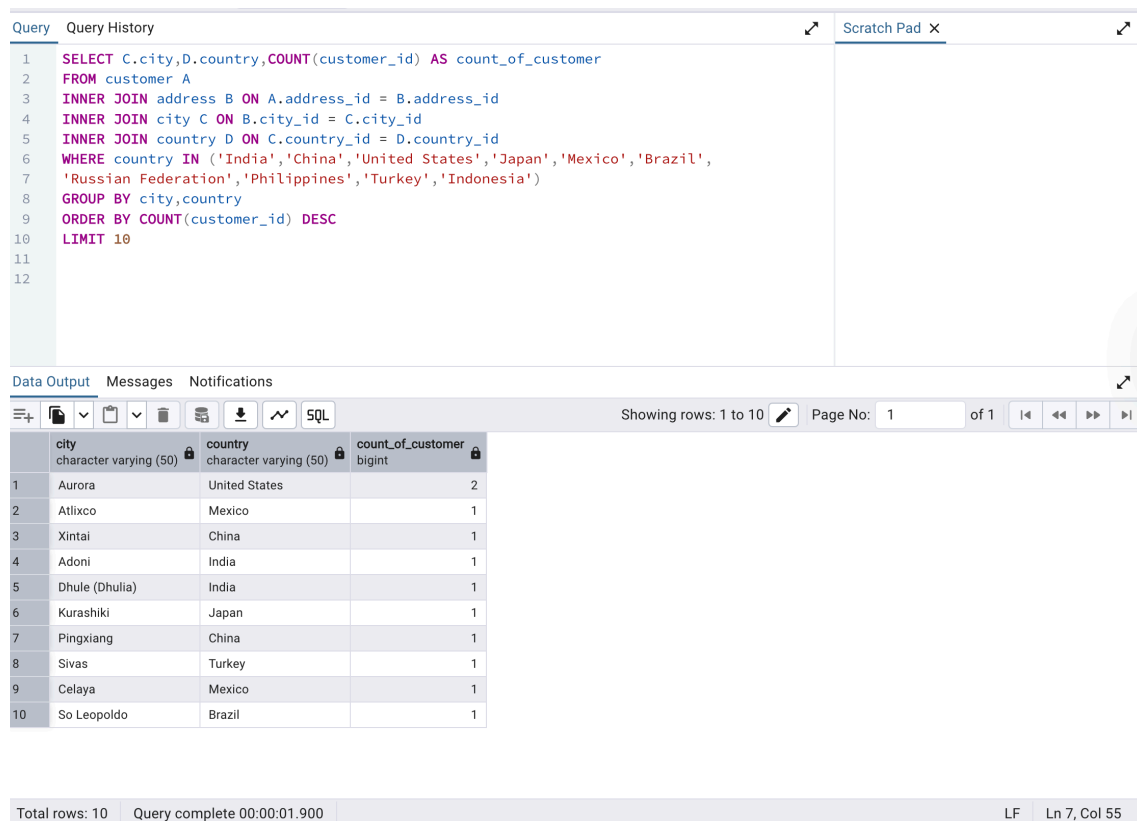
I have grouped the results by country since we wanted the top 10 countries where Rockbuster has the maximum customers.

Identifying the top 10 cities that fall within the top 10 countries I identified in step 1.

```

SELECT C.city,D.country,COUNT(customer_id) AS count_of_customer
FROM customer A
INNER JOIN address B ON A.address_id = B.address_id
INNER JOIN city C ON B.city_id = C.city_id
INNER JOIN country D ON C.country_id = D.country_id
WHERE country IN ('India','China','United States','Japan','Mexico','Brazil',
'Russian Federation','Philippines','Turkey','Indonesia')
GROUP BY city,country
ORDER BY COUNT(customer_id) DESC
LIMIT 10

```



The screenshot shows a SQL IDE interface. The top pane displays a SQL query that filters for 10 specific countries and orders cities by the number of customers. The bottom pane shows the results in a table with 10 rows.

Query:

```

1 SELECT C.city,D.country,COUNT(customer_id) AS count_of_customer
2 FROM customer A
3 INNER JOIN address B ON A.address_id = B.address_id
4 INNER JOIN city C ON B.city_id = C.city_id
5 INNER JOIN country D ON C.country_id = D.country_id
6 WHERE country IN ('India','China','United States','Japan','Mexico','Brazil',
7 'Russian Federation','Philippines','Turkey','Indonesia')
8 GROUP BY city,country
9 ORDER BY COUNT(customer_id) DESC
10 LIMIT 10

```

Data Output:

	city character varying (50)	country character varying (50)	count_of_customer bigint
1	Aurora	United States	2
2	Atlixco	Mexico	1
3	Xintai	China	1
4	Adoni	India	1
5	Dhule (Dhulia)	India	1
6	Kurashiki	Japan	1
7	Pingxiang	China	1
8	Sivas	Turkey	1
9	Celaya	Mexico	1
10	So Leopoldo	Brazil	1

Total rows: 10 Query complete 00:00:01.900 LF Ln 7, Col 55

Since I now know the top 10 countries, I put them in the WHERE clause. I also selected the city name along with the country name to display in my output. The other change I made in this query was to group it by both city and country.

Finding the top 5 customers from the top 10 cities who've paid the highest total amounts to Rockbuster. The customer team would like to reward them for their loyalty!

```

SELECT A.customer_id,
B.first_name,
B.last_name,
D.city,
E.country,
SUM(amount) AS total_amount_paid

```

```

FROM payment A
INNER JOIN customer B ON A.customer_id = B.customer_id
INNER JOIN address C ON B.address_id = C.address_id
INNER JOIN city D ON C.city_id = D.city_id
INNER JOIN country E ON D.country_id = E.country_id
WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
'Pingxiang','Sivas','Celaya','So Leopoldo')
AND country IN ('India','China','United States','Japan','Mexico','Brazil',
'Russian Federation','Philippines','Turkey','Indonesia')
GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
ORDER BY SUM(amount) DESC
LIMIT 5

```

Query

Query History

Scratch Pad

```
1 SELECT A.customer_id,
2 B.first_name,
3 B.last_name,
4 D.city,
5 E.country,
6 SUM(amount) AS total_amount_paid
7 FROM payment A
8 INNER JOIN customer B ON A.customer_id = B.customer_id
9 INNER JOIN address C ON B.address_id = C.address_id
10 INNER JOIN city D ON C.city_id = D.city_id
11 INNER JOIN country E ON D.country_id = E.country_id
12 WHERE city IN ('Aurora','Atlixco','Xintai','Adoni','Dhule(Dhulia)','Kurashiki',
13 'Pingxiang','Sivas','Celaya','So Leopoldo')
14 AND country IN ('India','China','United States','Japan','Mexico','Brazil',
15 'Russian Federation','Philippines','Turkey','Indonesia')
16 GROUP BY A.customer_id,B.first_name,B.last_name,D.city,E.country
17 ORDER BY SUM(amount) DESC
18 LIMIT 5
19
```

Data Output

Messages

Notifications

</

For this query, we have 5 tables (payment, customer, address, city and country). We'll be using two WHERE clauses specifying both the city and country.